MILITARY SPECIFICATION

FLASK, VOLUMETRIC, PLASTIC

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers one type of autoclavable plastic, volumetric flask with snap-on cap suitable for use in general laboratory application.

1.2 Classification. Volumetric flasks covered by this specification shall be of the following sizes, as specified (see 6.2):

Size 1. - 100 milliliters (ml) capacity
Size 2. - 250 milliliters (ml) capacity
Size 3. - 500 milliliters (ml) capacity
Size 4. - 1000 milliliters (ml) capacity

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

Commercial Standard (CS)

CS 21	Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical Ground Joints.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20401).

3. REQUIREMENTS

3.1 Material.

3.1.1 Polypropylene. Flasks shall be fabricated from a translucent, virgin polypropylene plastic material.

3.1.2 Polyethylene. Snap-on cap shall be fabricated from a flexible, translucent, conventional polyethylene.

3.2 Resistance to steam heat sterilization. Flasks shall show no evidence of deterioration when tested as specified in 4.4.1.

3.3 Design.

3.3.1 Flasks. Flasks shall be of one-piece construction, having a long straight neck, a rounded bulb, and a flat base. The neck shall be straight and perpendicular to the base and circular in cross section. The neck shall extend above and below the capacity mark (see 3.5). The inner surface at the top of the neck shall be tapered to accommodate standard size tapered stoppers

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(see Table I). The top of the neck shall have a lip on the outside rim, suitable for retaining a snap-on-type cap. The bottom of the neck shall begin to expand into a rounded bulb and the bottom of bulb shall be flat to serve as a base. The flask shall stand solidly on a level surface and shall also stand on a plane inclined 15° to the horizontal. Flasks shall have a spot suitable for marking located on the front of the bulb.

3.3.2 Snap-on cap. Snap-on cap shall fit snugly over the top of the flask. Top of cap may be recessed.

3.4 Dimensions, tolerances and standard stopper number. Dimensions shall be as specified in table I. The standard stopper numbers specified in table I shall be in accordance with CS 21. Capacity shall be within the tolerances specified in table I when tested as specified in 4.4.2.

Table I - Dimensions, Tolerances and Standard Stopper Size

Size	Capacity Tolerances (ml)	Overall Height and Tolerances (in)	Maximum outside diameter of Bulb (in)	Standard Stopper Number
1	100 ± 0.16	6-3/4 ± 1/4	2-1/4	13
2	250 ± 0.24	8-1/8 ± 1/4	3-1/4	13
3	500 ± 0.40	9-3/4 ± 1/4	3-7/8	16
4	1000 ± 0.60	11-1/2 ± 1/4	4-3/4	19

3.5 Markings.

3.5.1 Capacity mark. Each flask shall be provided with a capacity line which shall be fine, clean, permanent, continuous and of uniform width, perpendicular to the axis and parallel to the base of the flask. The capacity mark shall be inscribed on the neck of the flask. Flasks shall be calibrated to contain "TC" and to deliver "TD" at 25° C. within the tolerances specified in table I.

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3.5.2 Identification marking. Flasks shall be permanently and legibly marked with the manufacturer's name or trademark, the capacity, the temperature at which flasks were calibrated and the symbols "TC/TD".

3.6 Workmanship. Workmanship shall be first class throughout. Volumetric flasks and caps shall be free from defects which detract from their appearance or may impair their serviceability. Flasks shall be smoothly finished and free from blisters, blooms, orange-peel, cracks and discoloration. Caps shall be free from mottling, discoloration, cracks, chip and blisters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination purposes shall be the flask and cap.

Table II. Sampling for Examination

	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for testing purposes shall be the flask.

4.3 Examination. Flask and caps shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with Table III. The flask and caps shall be examined for defects but shall not be restricted to the classified possible defects.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Table III.

Categories and defects

Major	
101	Neck of flask not perpendicular to a horizontal plane.
102	Flasks and caps not uniform in thickness, color, texture or finish.
103	Flasks not free from checks, cracks, chips, blisters or scratches.
104	Base not perpendicular to a vertical axis.
105	Capacity mark or identification markings illegible, incorrect, or incomplete.
106	Snap-on caps fail to fit snugly.
107	Edges not smooth or free from protrusions, burs, or sharp edges.

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Table III. (cont'd.)

Categories and defects

Major	
108	Neck of flask fails to accommodate required standard stopper (see 3.6 for the required stopper size).
109	Snap-on caps not furnished with flasks.
Minor	
201	Flasks and caps not free from dirt, grease or other foreign matter.

4.4 Sampling for tests. Test shall be conducted to determine compliance with specification requirements. Units of product for test purposes shall be as specified in Table IV.

Table IV - Sampling for Tests

Unit of Product	Characteristic	Requirement	Test Procedure	Inspection level
Flask	Autoclavability	3.2	4.4.1	S-2
Flask	Capacity	3.4	4.4.2	S-2

NOTE: After autoclaving, capacities and tolerances shall be in accordance with requirements specified in Table I.

4.4.1 Resistance to steam heat. Flasks to be tested shall be placed in an autoclave that is equipped with controls for maintaining and measuring temperature and pressure during the autoclaving period. Flasks shall be exposed to steam at a temperature of 121° ± 2° Centigrade (C.) or 249.8° ± 3.6° Fahrenheit (F.) for 20 ± 1/2 minutes. At the end of the exposure period, the steam shall be blown off and the flasks removed from the autoclave, within 3 minutes. If any wax or bloom is present on the surfaces of the flasks after the exposure period, the flasks shall be scrubbed with soap, water and a soft brush to remove it. The flasks shall then be dried in air out of direct light. The exposure and drying shall be repeated until a total of 5 such exposures have been completed. Flasks shall be allowed to stand at least 20 minutes between exposures, from the time it is removed from the steam, before repeating the operation. At the end of the fifth exposure period, the steam shall be blown off, and the flasks removed from the autoclave. The flasks shall then be dried in air out of direct light for not less than 16 hours. Examine flasks for deterioration.

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4.4.2 Accuracy. Flasks shall be thoroughly clean, and rinsed using distilled water. Test shall be conducted at 25° ± 2° C. Flasks and water shall be allowed to stand at test temperature for not less than 30 minutes before starting the test. The empty flask shall be weighed; the weight recorded, and the temperature of the water shall be recorded. Then carefully fill the flask to its capacity mark with distilled water. Any excess water above the capacity mark or on the outside of the flask shall be removed with absorbent paper. Weigh the filled flask and record the weight. Obtain the relative density of the water, corresponding to the observed temperature. The volume of the flask may be calculated by using the following formula:

$$V = \frac{M_2 - M_1}{D}$$

Where:

V = Volume in milliliters (ml) of flask.
M_2 = Weight in grams (gm) of the filled flask.
M_1 = Weight in grams (gm) of the empty flask.
D = Absolute Density of Water at observed temperature.

After the volume of the flask has been determined, the water shall be transferred to another thoroughly cleaned, tare-weighted flask. The flask shall be allowed to drain for not less than 10 seconds. The volume of water in the new flask shall be determined in the same matter as that of the first. The delivered volume shall be within the tolerances specified in table I.

NOTE: Weighings must be corrected for the bouyancy of air.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One flask, of the size specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Size	Unit package	Intermediate package	Exterior container
1	1 unit	12 units	144 units
2	1 unit	12 units	48 units
3	1 unit	8 units	48 units
4	1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each flask shall be thoroughly clean and packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Volumetric flasks covered by this specification are intended for use in general laboratory procedures for which steam sterilization may be required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Federal stock number (FSN).
(c) Specify size flask required (see 1.2).
(d) Selection of applicable levels of packing. (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.4 This specification covers the following items shown in the Federal Supply Catalog:

FSN	Item Identification
6640-938-4723	FLASK, VOLUMETRIC, Plastic, 100 ml
6640-938-4758	FLASK, VOLUMETRIC, Plastic, 250 ml
6640-938-4763	FLASK, VOLUMETRIC, Plastic, 500 ml
6640-938-4775	FLASK, VOLUMETRIC, Plastic, 1000 ml

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0650

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R755

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: MIL-P-36676(DSA-DM) PSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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